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Research Paper

Investigating cultural influences on the associations between rumination and symptoms of posttraumatic stress disorder among European Australian and Chinese Australian trauma survivors

James Haoxiang Li[a,*], Larissa Shiying Qiu[a], Joshua Wong[b], Winnie Lau[c], Richard A. Bryant[b], July Lies[a], Belinda J. Liddell[d], Laura Jobson[a]

[a] *Turner Institute for Brain and Mental Health and School of Psychological Sciences, Monash University, Clayton, VIC, Australia*
[b] *School of Psychology, University of New South Wales, Kensington, Sydney, NSW, Australia*
[c] *Phoenix Australia-Centre for Posttraumatic Mental Health and Department of Psychiatry, University of Melbourne, Carlton, VIC, Australia*
[d] *School of Psychological Sciences, University of Newcastle, Callaghan, NSW, Australia*

A R T I C L E I N F O

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A B S T R A C T

This study investigated the moderating effect of cultural variables (cultural group, self-construal and cognitive style) on the association between rumination and PTSD symptoms among European Australian and Chinese Australian adult trauma survivors. European Australian (n = 111) and Chinese Australian (n = 111) trauma survivors were recruited through social media advertisements on Facebook, WeChat and RED and completed an on-line questionnaire assessing rumination, cultural variables and PTSD symptomatology. There was no evidence that cultural group moderated the associations between brooding and PTSD symptomatology. interdependent self-construal moderated the associations between all three types of rumination and PTSD symptomatology, such that the association between each type of rumination (brooding, reflection and trauma-specific) and PTSD symptomatology increased at higher levels of interdependence. These findings highlight the importance of considering rumination regardless of the cultural background of trauma survivors. Additionally, it may be clinically beneficial to consider the impacts of trauma and rumination on interdependent aspects of the self and consider the socio-cultural context (including an individual's values and self-concept). Further research is crucial for integrate cross-cultural psychology theories and empirical research to better conceptualize rumination and its role in maintaining psychopathology in Asian contexts.

Posttraumatic stress disorder (PTSD) is a debilitating psychological condition that has been observed in most societies (Hinton and Lewis-Fernández, 2011). Despite observations across societies, little is still understood about how cultural factors may affect the development and maintenance of the disorder. One key factor that has received considerable attention with respect to PTSD is rumination, a cognitive process characterized by repetitively dwelling on one's negative emotions and their causes and consequences (Nolen-Hoeksema, 1991). While there have been significant theoretical and empirical advancements implicating rumination as a key maintenance process in PTSD (Moulds et al., 2020; Seligowski et al., 2015), a significant limitation of this research is the predominance of sampling Western participants. Thus, trauma survivors from culturally diverse backgrounds are under-represented. This is problematic as research has shown that psychological treatment effects improve significantly when culturally understood and tailored (Huey and Tilley, 2018). To date, there is little empirical research to guide this tailoring, highlighting the urgent need for more cross-cultural research in this area.

While research has consistently shown that rumination contributes to the development and maintenance of PTSD (Moulds, 2020; Seligowski et al., 2015), three key critiques of rumination research in PTSD have emerged. First, researchers have distinguished between different forms of rumination. For instance, a distinction is made between brooding (i.e., a passive preoccupation with negative aspects of oneself or negative interpretations of one's circumstances) and reflection (i.e., purposeful engagement in cognitive problem-solving, motivated by self-curiosity) (Trapnell & Campbell, 1999). Studies have found that while brooding is consistently associated with elevated PTSD symptoms

* Corresponding author.
E-mail address: james@stia.com.au (J.H. Li).

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(Cox et al., 2012; Moulds et al., 2020), the relationship between reflection and well-being is less clear (Cox et al., 2012; Treynor et al., 2003) and requires further investigation in the area of trauma.

Second, clinical research examining rumination has predominantly occurred in the domain of depression (Moulds et al., 2020). However, rumination is a transdiagnostic cognitive process (Hernández-Posadas et al., 2024; Wong et al., 2023). Thus, in the context of trauma, it is necessary to examine rumination that is explicitly trauma-related (Moulds et al., 2020). This argument follows theoretical accounts that explicitly implicate trauma-related rumination, rather than general rumination, in the maintenance of PTSD (Ehlers and Clark, 2000). Thus, rumination is a multidimensional construct, requiring consideration of different forms of rumination (including brooding, reflection and trauma-specific) in order to draw more meaningful conclusions in PTSD research. Unfortunately, extant studies examining rumination in trauma survivors focus solely on brooding rumination.

Third, studies investigating rumination in the context of trauma have predominantly utilized Western trauma survivor samples (Seligowski et al., 2015). There is a paucity of cross-cultural research in this area and a pervasive assumption that rumination operates in universally similar ways for all trauma survivors (Nagulendran and Jobson, 2020). Emerging cross-cultural paradigms, theories and empirical studies have challenged these assumptions and suggest that rumination may differ across cultures. Several empirical studies have found that although individuals with East-Asian cultural heritage tend to ruminate more than those with Western European cultural heritage, rumination tends to be less predictive of psychopathology and poor wellbeing in East Asian cultural groups (e.g., Chang et al., 2010; Jobson et al., 2022; Kwon et al., 2013; Toussaint et al., 2023). Simultaneously, cross-cultural researchers have identified key dimensions by which cultural groups systematically vary, including self-construal, cognitive style and view of emotions (Zhang and Cross, 2011; de Vaus et al., 2018), which are believed to explain cultural differences in how rumination relates to psychological adjustment.

Cultures tend to differ in the conceptualisation of the self in relation to others (self-construal). Western individualistic cultures (e.g., Australia, US) tend to hold an independent self-construal, whereby the self is conceived as unique, independent, and detached from the social context (Markus and Kitayama, 2010). In contrast, East Asian collectivistic cultures (e.g., China, Japan) tend to adopt an interdependent self-construal, where one's interconnectedness and harmony with others is emphasized to a greater degree (Markus and Kitayama, 2010). Cultures can also vary based on cognitive style. Western individualistic cultures tend to endorse *analytic thinking styles*, characterized by distinguishing focal points from their contexts, and ascribing causality to events based on formal logic (Nisbett et al., 2010). In contrast, East-Asian cultures tend to adopt a *holistic* thinking style, characterized by paying attention to relationships between objects and their contexts (Nisbett et al., 2010).

These frameworks may shape rumination. Cultural groups that are likely to have independent self-construal and analytic thinking styles (i.e., Western cultures) may be more likely to ruminate from a self-immersed perspective and fixate on the content of their own negative emotions (de Vaus et al., 2018; Grossman and Kross, 2010). These individuals may also be inclined to understand negative emotions as caused by their internal qualities, rather than external circumstances, thus evoking greater distress (de Vaus et al., 2018). These patterns are contrasted with other cultural groups that are likely to have interdependent self-construal and holistic thinking styles (i.e., Asian cultures). Hence, rumination in these cultures may be a more self-distanced process, whereby individuals view their experiences with a broader focus on social and contextual factors (de Vaus et al., 2018). This, in turn may facilitate adaptively gaining insight and closure over persistent negative thought patterns, rather than re-experiencing of negative emotion (see de Vaus et al., 2018 for full review).

Despite compelling theory and evidence suggesting that rumination may differ across cultures, cultural differences in rumination have rarely been explored in the context of PTSD. Moreover, cross-cultural studies in this area often use broad cultural grouping as a basis of comparison. However, research indicates it is not simply cultural group membership that influences cognitive and emotion regulation processes and their association with mental health, but rather an individual's orientation towards particular cultural values or cognitive style (e.g., independence versus interdependence and holistic versus analytic cognitive style) (Ford and Mauss, 2015). Therefore, it is crucial to investigate the relationships between different types of rumination and PTSD symptoms, and how cultural variables may influence this relationship.

This study aimed to investigate the moderating effect of cultural variables on the relationship between rumination and PTSD symptoms among Australian (with European heritage) and Chinese (with East Asian heritage) adult trauma survivors. We hypothesized that rumination (brooding, reflection and trauma-specific) would be associated with higher levels of PTSD symptoms in the overall sample (Hypothesis 1). Second, we hypothesized that cultural variables (cultural group, self-construal, holistic cognitive style) would moderate these associations, such that the associations between higher levels of trauma-specific and brooding rumination and greater PTSD symptoms would be stronger for (a) European Australian trauma survivors than Chinese Australian trauma survivors (Hypothesis 2a), (b) those with greater levels of independent self-construal (Hypothesis 2b), (c) those with lower levels of interdependent self-construal (Hypothesis 2c) and (d), those with lower holistic cognitive style (Hypothesis 2d). Given the associations between reflective rumination and mental health are less clear (Cox et al., 2012; Treynor et al., 2003), in the current study these associations remained exploratory. We also conducted exploratory analyses to examine whether cultural group interacted with self-construal or holistic thinking style to moderate the associations between each type of rumination and PTSD symptoms.

1. Method

1.1. Design

This study employed a cross-sectional correlational design with two cultural groups: European Australian group and Chinese Australian group residing in Australia. Ethical approval was obtained from BLINDED (32,521). This study was part of a larger research project, BLINDED, and unrelated aspects of this project have been published elsewhere (BLINDED) (Jobson et al., 2024).

1.2. Participants

Using convenience sampling, participants were recruited from the Australian general community via online social media advertisements on Facebook, WeChat and RED (a Chinese social media and e-commerce platform). Eligibility criteria included a) having experienced at least one DSM-5 PTSD Criterion A traumatic event, as indicated by the Life Events Checklist-5 (Weathers et al., 2013), b) being between 18 and 65 years of age, c) either born in Australia and culturally self-identifying as Western European or born in China and having moved to Australia in the last 15 years, and culturally self-identifying as Chinese, d) having all four grandparents of the same cultural background as the respondent, and e) able to complete the survey in English or simplified Chinese. We restricted Chinese participants to those who had migrated within the past 15 years to minimize potential confounding effects of acculturation (Schwartz et al., 2010).

The survey was initially completed by 388 respondents, and their responses were assessed for quality. Responses were excluded if (a) the survey was completed in <15 min, (b) the reCAPTCHA score was below 0.5 s (Mancenido et al., 2021), c) the Conscientious Responder Scale score was < 3 (Marjanovic et al., 2014), and d) the index trauma did not meet Criterion A, the research standard for traumatic events (e.g., life-threatening experiences, per the Life Events Checklist; Weathers et al., 2013). For the final criterion, cases that did not clearly fit any trauma category were reviewed individually by researchers and then discussed collectively, resulting in exclusion of 13 participants. Based on these exclusion criteria, 119 responses were excluded leaving 266 participants for analysis. Additionally, as there was a significant difference in measured PTSD symptoms between the groups (57 % of the European Australian group met the cut-off for provisional PTSD diagnosis versus 33.3 % of the Chinese Australian group). To address this potentially confounding imbalance, we utilised case-matching, a standard method to equate groups on key variables (Austin, 2011; Stuart, 2010), to control for group differences in PTSD symptoms (see Data Analysis Section). This left a final sample of 222 participants in the final sample (111 per cultural group).

2. Materials

2.1. Life events checklist-5 (LEC)

The LEC-5 (Weathers et al., 2013) was used to screen for lifetime exposure to Criterion A traumatic events. Participants self-reported their experiences with 17 types of potentially traumatic events on 6-point nominal scales, and then nominated their index trauma (i.e., the single worst event), including brief details about the event and when it occurred (Weathers et al., 2013). The LEC-5 is not scored, but rather used to confirm trauma exposure (i.e., eligibility criteria), categorize participants' index traumas and calculate time since trauma.

2.2. PTSD checklist for DSM-5 (PCL-5)

The PCL-5 (Weathers et al., 2013) was used to assess PTSD symptom severity. Participants completed the 20 items of this scale in relation to their index trauma (as indicated in LEC-5). Items are self-rated using a 5-point Likert scale (0 = *not at all*, 4 = *extremely*). Scores are summed to yield a total score ranging from 0–80, with higher scores indicating greater PTSD symptom severity and a score greater than 33 being indicative of probable PTSD (Weathers et al., 2013). The PCL-5 has demonstrated good psychometric properties, including in cross cultural research (Itoh et al., 2017), and in Chinese samples (Fung et al., 2019). In the present study, this measure had excellent internal consistency (European Australian group $\alpha = 0.94$, Chinese Australian group $\alpha = 0.94$).

2.3. Ruminative response scale-short form (RRS-SF)

The RRS-SF (Nolen-Hoeksema and Morrow, 1991) was used to measure brooding and reflection rumination. It consists of two subscales: brooding (5 items, e.g., "I think why I can't handle things better?") and reflection (5 items, e.g., "write down what you are thinking, and analyze it"). Participants self-rated the ten items along a 4-point Likert-type how often they engage in each type of rumination when experiencing negative emotions (1 = *almost never*, 4 = *almost always*). Summed scores range from 5 to 20 for each subscale, with higher scores indicating higher levels of the corresponding type of rumination. The RRS-SF has demonstrated good psychometric properties, including in cross cultural research involving Chinese participants (He et al., 2021). In the present study, this measure had good internal consistency for both brooding (European Australian group $\alpha = 0.76$, Chinese Australian group $\alpha = 0.72$) and reflection (European Australian group $\alpha = 0.74$, Chinese Australian group $\alpha = 0.79$) subscales.

2.4. Repetitive thinking questionnaire-10 (RTQ-10)

The RTQ-10 (McEvoy et al., 2010) was used to measure trauma-specific rumination. Participants completed the RTQ-10 in relation to the index trauma indicated on the LEC-5. The scale comprised 10 items (e.g., "I think about the situation all the time") which were self-rated using a 5-point Likert scale (1 = *not at all*, 5 = *very true*). Scores were summed to yield a total score between 10–50, with higher scores indicating greater trauma-specific rumination. The RTQ-10 has demonstrated good psychometric properties, including in cross-cultural research involving Chinese participants (Chang et al., 2010). In this study, internal consistency was excellent (European Australian group α = 0.91, Chinese Australian group α = 0.89).

2.5. Self-Construal scale (SCS)

The SCS (Singelis, 1994) is a 30-item self-report measure of how people define themselves in relation to others. It consists of two subscales: independent (15 items; e.g., "I feel it is important for me to act as an independent person") and interdependent (15 items; e.g., "It is important for me to maintain harmony within my group"). Participants rated their agreement with each statement on a 7-point Likert scale (1= *strongly disagree*, 7= *strongly agree*) and scores were summed for both subscales. The SCS has been frequently used in previous cross-cultural research and has been validated within Chinese samples (Feng et al., 2022; Wei et al., 2022). In this study, the SCS showed acceptable internal consistency for independent (European Australian group α = 0.81; Chinese Australian group α = 0.76) and interdependent self-construal (European Australian group α = 0.80, Chinese Australian group α = 0.78).

2.6. Holistic cognitions scale (HCS)

The HCS (Lux et al., 2021) is a 16-item self-report measure of holistic and analytic thinking styles. Items in the HCS are divided into four constructs (four items each) that differ between holistic and analytic thinking: attention, causality (reverse scored), contradiction, and perception of change (reverse scored). Participants rated their agreement with each item (e.g., "events occur as a result of individuals' choices and behaviour") on a 7-point Likert scale (1= *completely disagree*, 7= *completely agree*) and scores were summed for both the total measure and for each subscale. Higher sum scores indicated greater holistic thinking (and less analytic thinking). In the present study, the HCS showed acceptable to moderate internal consistency (European Australian group α = 0.60, Chinese Australian group α = 0.69).

2.7. Conscientious responders scale (CRS)

The CRS (Marjanovic et al., 2014) was used as a validity check to identify and exclude participants who answered questions hastily and inaccurately. The five items were dispersed throughout the online questionnaire. Each item instructs participants to respond in a certain way (e.g., "*to respond to this question, please choose option number five, 'slightly agree'*"). Participants' responses are scored as either correct (1) or incorrect (0). Scores ranged from 0 to 5, and participants who scored <3 on the CRS were excluded from data analyses, based on slim chance of random responses achieving a score of 3 or above (<3 %) (Marjanovic et al., 2014).

3. Demographic survey

Participants were asked to self-report their age, gender, highest level of education attained, cultural background, and religion at the end of the survey.

4. Procedure

To ensure cultural appropriateness, all English measures were translated into Chinese and were then back-translated into English by bilingual researchers using gold-standard translation approaches (Gorecki et al., 2014). Translated copies were assessed by bilingual psychologists to ensure validity. The study was advertised on social media. Interested respondents contacted the researchers via email and were screened via email before receiving a link to the online questionnaire on Qualtrics. Participants completed the above measures and were then reimbursed with a $25 digital gift-voucher.

5. Data analyses

Statistical analyses were conducted using SPSS version 27.0 (IBM, 2024). As noted above, there was a significant difference in PTSD symptoms between the two cultural groups (European Australian group M= 34.98, SD = 18.57, Chinese Australian M= 26.96 SD= 16.73. Therefore, we utilized case-control matching. We matched European Australian participants to Chinese Australian participants based on their PCL-5 total scores. We used a match tolerance of 5. This led to 111 matches (3 exact matches, 108 'fuzzy' matches) and 44 participants did not have a match and were excluded from the analyses. Thus, our data analyses were conducted on a final sample of 222 participants (111 in each cultural group). All analyses presented in the Results section is based on the case-control matched sample. It is worth noting that when we used the full dataset to conduct the analyses, a similar pattern of results emerged to that reported below. Initial checking did not identify any missing data. Examination of Z-scores revealed 14 univariate outliers across multiple variables, which were subsequently Winsorized by replacing them with the observations closest to them (Beaumont and Rivest, 2009). As some variables did not meet assumptions for normality, bootstrapping was utilized for all analyses. As shown below, several group differences were found in age, gender, education, religion, index trauma, and time since trauma. Thus, we included these variables as covariates in all analyses.

To examine Hypothesis 1, partial correlation analysis was used to examine the relationships between rumination (brooding, reflection, trauma-specific) and PTSD symptomatology, controlling for age, gender, education, religion, time since trauma and index trauma. To examine Hypothesis 2, moderation analyses were conducted using PROCESS v2.4 (Model 1) with 5000 bootstrap resamples. Moderation analyses were conducted separately for brooding, reflection and trauma-specific rumination. Significance was indicated by the confidence interval not including zero. To assess our exploratory hypothesis, moderation analyses were conducted using PROCESS (Model 3). We entered cultural group and each cultural variable (independent self-construal, interdependent self-construal and holistic cognitive style) simultaneously as moderators to test their interaction with each type of rumination in their association with PTSD symptoms (3 cultural variables x 3 types of rumination = 9 models tested). To control for Type I error inflation due to multiple comparisons, we applied a Bonferroni adjustment, setting the significance threshold at $\alpha = 0.006$. For the moderation analyses, as self-construal and holism were continuous variables, the moderating effects of each of these variables were considered at low levels (1SD below the mean), medium levels (mean) and high levels (1SD above the mean).

6. Results

Table 1 presents participant characteristics and shows significant group differences in age (the European Australian group was older), gender (the European Australian group had more non-binary participants), education, religion, time since trauma (Chinese Australian group had more recent trauma) and trauma type. Over 30 % of each cultural group reported a PCL-5 score greater than the clinical cutoff score of 33, indicative of probable PTSD (European Australian group $n = 34$, Chinese Australian group $n = 34$). As expected, the Chinese Australian group reported significantly higher levels of interdependent self-construal than the European Australian group. The Chinese Australian group also reported significantly greater holistic cognitive style than the European Australian group (see Table 1). Additionally, as shown in Table 1, the European Australian group reported significantly greater brooding, reflection and trauma-related rumination than the Chinese Australian group.

Table 1
Descriptive statistics and group comparisons of characteristics (European Australian group versus Chinese group).

	European Australian ($n = 111$)	Chinese ($n = 111$)	Statistic
Age	36.20 (12.31)	29.48 (7.78)	$t(185) = 4.86^{***}$
Gender[a]	18:78:15	26:83:2	χ^2 (2, 222)= 11.56**
Education[b]	16:14:40:36:5	11:5:36:59:0	χ^2 (4, 222) = 15.97**
Religion[c]	50:48:0:12:0	79:11:11:2:8	χ^2 (4, 222) = 53.31***
Index trauma[d]	13:30:26:41:1	21:42:8:28:12	χ^2 (4, 222) = 25.17***
Time since trauma	13.74 (11.70)	8.02 (7.40)	$t(185) = 4.35^{***}$
PTSD symptoms	29.84 (15.72)	28.16 (16.34)	$t(220) = 0.78$
Brooding rumination	12.78 (3.38)	11.90 (2.94)	$t(220) = 2.08^{*}$
Reflection rumination	12.89 (3.43)	11.37 (3.14)	$t(220) = 3.45^{***}$
Trauma-specific rumination	28.83 (9.41)	26.06 (9.01)	$t(220) = 2.24^{*}$
Independent	70.46 (12.68)	67.56 (11.12)	$t(220) = 1.81$
Interdependent	64.05 (11.43)	68.66 (10.82)	$t(220) = 3.09^{**}$
Holistic thinking	73.70 (8.16)	77.00 (9.06)	$t(220) = 2.85^{**}$

Note: Data is displayed as mean (standard deviation) for all variables except gender, education, region and index trauma. Age and time since trauma are measured in years.
* $p < 0.05$.
** $p < 0.01$.
*** $p < .001$.
[a] Displayed as men: women: non-binary.
[b] Self-reported highest level of education attained, displayed as secondary education: post-secondary education: undergraduate degree: postgraduate degree: other: prefer not to say.
[c] Self-reported religious background, displayed as no religion: Christianity: Buddhism: other (including Agnosticism, Judaism, Islam and Paganism): did not report religion.
[d] Index trauma category, displayed as accident: non-sexual assault or abuse: sexual assault: witnessing death, serious injury or illness: war or natural disaster: other.

Hypothesis 1: Associations between rumination and PTSD symptomatology

Statistically significant moderate positive correlations were found between PTSD symptomatology and brooding $r = 0.36$, 95 % CI [.25, 0.48], $p < .001$, and reflection $r = 0.20$, 95 % CI [.06, 0.33], $p < .001$. A positive correlation was found between PTSD symptomatology and trauma-specific rumination ($r = 0.50$, 95 % CI [.38, 0.60], $p < .001$); see Supplementary Material for correlation analyses conducted separately for each cultural group, with both groups showing similar correlation patterns.

Hypothesis 2: Moderation analyses

Hypothesis 2a: Cultural group as a moderator

Table 2
Model results of whether cultural group moderated the relationship between rumination types and PTSD symptoms.

	B (SE)	95 %CI	p
PTSD symptoms			
Brooding x Cultural group	.83(0.64)	−0.44 – 2.11	0.20
Reflection x Cultural group	.81(0.66)	−0.49 – 2.10	0.22
Trauma rumination x Cultural group	.26(0.21)	−0.15 – 0.67	0.21

As shown in Table 2, there was no evidence that cultural group moderated the associations between brooding and PTSD symptomatology, $R^2\ \Delta = 0.01$, $F(1, 212) = 1.65$, $p = .20$, reflection and PTSD symptomatology, $R^2\ \Delta = 0.01$, $F(1, 212) = 1.50$, $p = 0.22$, or trauma-specific rumination and PTSD symptomatology, $R^2\ \Delta = 0.01$, $F(1, 212) = 1.58$, $p = 0.21$.

Hypothesis 2b: Independent self-construal as a moderator

As shown in Table 3, there was no evidence that independent self-construal moderated the associations between any type of rumination and PTSD symptomatology.

Hypothesis 2c: Interdependent self-construal as a moderator

As shown in Table 3, interdependent self-construal moderated the associations between all three types of rumination and PTSD symptomatology. Follow-up analyses showed that for trauma survivors with medium ($B = 1.70$, $SE= 0.32$, 95 %[1.07, 2.32], $p< .001$), and high interdependent self-construal ($B = 2.61$, $SE= 0.44$, 95 %[1.75, 3.48], $p<$ .001), greater brooding was significantly associated with higher levels of PTSD symptoms; this association was not significant for those with low levels of interdependent self-construal ($B = 0.83$, $SE= 0.51$, 95 % [−0.17, 1.83], $p = 0.10$). Similarly, for trauma survivors with medium interdependent self-construal ($B = 1.00$, $SE= 0.31$, 95 %[.38, 1.62], $p =$ .002), and high interdependent self-construal ($B = 1.69$, $SE= 0.47$, 95 % [.77, 2.61], $p< .001$), greater reflection was significantly associated with higher PTSD symptomology; this association was not significant for those with low levels of interdependent self-construal ($B = 0.35$, $SE=$ 0.41, 95 % [−0.47, 1.17], $p = 0.40$). Finally, for all individuals, regardless of level of interdependent self-construal, trauma-specific rumination was associated with higher PTSD symptoms (low interdependent self-construal: $B=$. 60, $SE= 0.15$, 95 %[.31, 0.91], $p< .001$, medium interdependent self-construal: $B = 0.84$, $SE= 0.10$, 95 %[.64, 0.1.04], $p< .001$, high interdependent self-construal: $B = 1.09$, $SE=$ 0.14, 95 %[.82, 1.36], $p< .001$). The strength of the association between each type of rumination (brooding, reflection and trauma-specific) and PTSD symptomatology increased at higher levels of interdependence. These interaction effects are shown in Fig. 1.

Hypothesis 2d: Holistic cognitive style as a moderator

As shown in Table 3, holistic cognitive style did not moderate the associations between any type of rumination and PTSD.

6.1. Exploratory analyses

When cultural group and self-construal were entered simultaneously as moderators, there was no significant interaction found to moderate the associations between any type of rumination and PTSD. When cultural group and holistic thinking style were entered simultaneously as moderators, there was no significant interaction found to moderate the associations between any type of rumination and PTSD. Regression coefficients for these analyses are displayed in Table 4.

Table 3
Model results of whether cultural variables moderated the relationship between rumination types and PTSD symptoms.

	B (*SE*)	95 %CI	*p*
PTSD symptoms			
Brooding x Independence	−0.03(0.03)	−0.08–0.02	0.17
Reflection x Independence	−0.03(0.03)	−0.08–0.02	0.22
Trauma rumination x Independence	−0.01(0.01)	−0.01–0.01	0.37
PTSD symptoms			
Brooding x Interdependence	08(0.03)	0.02 – 0.14	0.01*
Reflection x Interdependence	06(0.03)	0.01–0.11	0.03*
Trauma rumination x Interdependence	02(0.01)	0.00–0.04	0.02*
PTSD symptoms			
Brooding x Holism	−0.01(0.03)	−0.08–0.07	.96
Reflection x Holism	−0.01(0.04)	−0.08–0.07	0.88
Trauma rumination x Holism	01(0.01)	−0.01–0.03	0.39

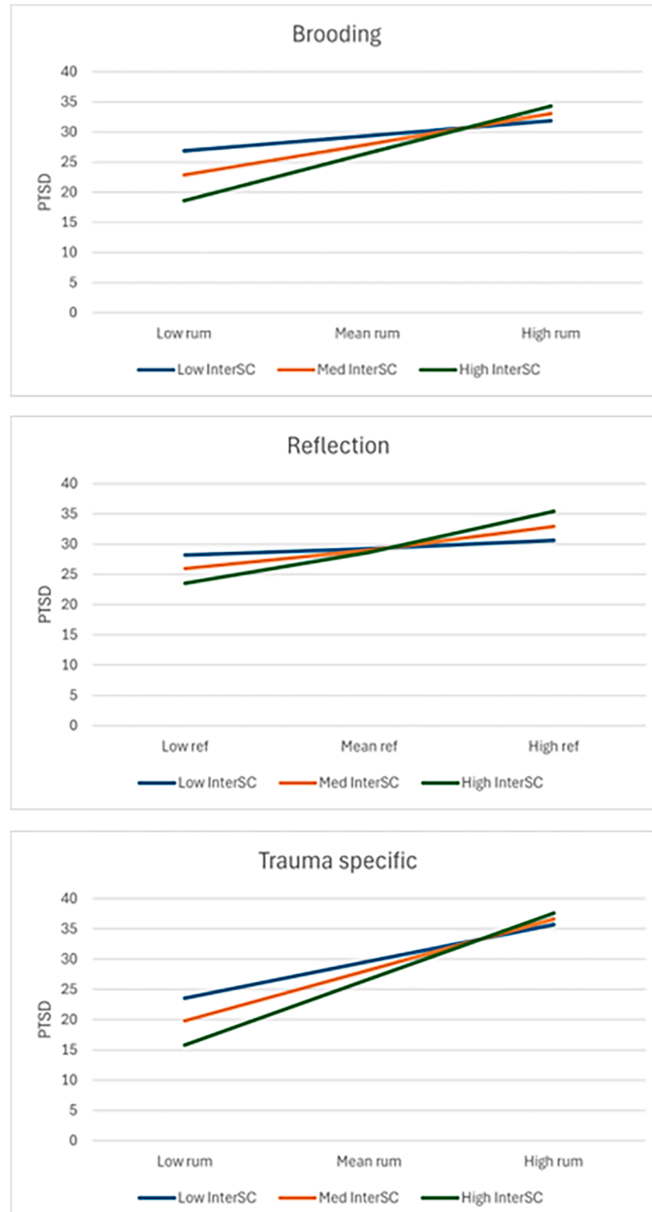

Fig. 1. The moderating effect of interdependent self-construal on the relationship between rumination and PTSD symptoms.
Note: InterSC – Interdependent self-construal.

7. Discussion

This study investigated the moderating effect of cultural variables (cultural group, self-construal and holistic thinking style) on the relationship between different types of rumination and PTSD symptoms. In support of Hypothesis 1, we found that rumination was associated with PTSD symptoms. Contrary to Hypothesis 2a, there was no evidence that cultural group moderated the positive associations between any type of rumination and PTSD symptoms. There was also no evidence to support Hypothesis 2b; independent self-construal did not moderate the associations between any type of rumination and PTSD symptoms. Regarding Hypothesis 2c, we found that interdependent self-construal moderated the associations between all three types of rumination and PTSD symptoms. However, contrary to our predictions, the strength of the association between each type of rumination and PTSD symptomatology increased at higher levels of interdependence. Contrary to Hypothesis 2d, there was no evidence that holistic thinking style moderated the

Table 4
Model results of whether cultural group and cultural variables moderated the relationship between rumination types and PTSD symptoms.

	B (*SE*)	95 %CI	*p*
PTSD symptoms			
Brooding x Independence x Culture	02(0.05)	−0.08–0.13	0.65
Reflection x Independence x Culture	−0.04 (0.06)	−0.15–0.07	0.44
Trauma rumination x Independence x Culture	01(0.02)	−0.03–0.04	0.74
PTSD symptoms			
Brooding x Interdependence x Culture	−0.01 (0.06)	−0.14–0.11	0.83
Reflection x Interdependence x Culture	02(0.06)	−0.09–0.14	0.67
Trauma rumination x Interdependence x Culture	01(0.02)	−0.02–0.05	0.51
PTSD symptoms			
Brooding x Holism x Culture	−0.10 (0.08)	−0.26–0.05	.20
Reflection x Holism x Culture	−0.04 (0.07)	−0.19–0.10	0.56
Trauma rumination x Holism x Culture	−0.01 (0.02)	−0.06–0.04	0.75

associations between the three types of rumination and PTSD symptomatology. Regarding our exploratory aim, we found no evidence that cultural group interacted with either self-construal or holistic thinking style in moderating the associations between any type of rumination and PTSD symptoms.

Our findings that rumination was associated with PTSD symptoms align with previous research implicating rumination as a key maintenance factor in PTSD (Seligowski et al., 2015). However, there was no evidence of a moderating effect of cultural group on the relationship between any type of rumination and PTSD symptoms. This observation does not align with expected cross-cultural differences theorized and observed in previous non-clinical research (Chang et al., 2010; de Vaus et al., 2018). Given that our sample comprised trauma survivors with PTSD symptoms, it is possible that when considering psychopathology, rumination may contribute to greater PTSD symptomatology regardless of one's cultural background.

There was some evidence to support the idea that self-construal influenced the associations between rumination and PTSD symptoms. A significant moderating effect was found for interdependent self-construal. However, contrary to our predictions, the relationship between each type of rumination and PTSD symptoms was stronger at higher levels of interdependence. This is in contrast to arguments in the cross-cultural literature and previous non-trauma research that together suggest rumination may be less detrimental for individuals valuing interdependence (de Vaus et al., 2018, Grossman and Kross, 2010). It is possible that those valuing interdependent self-construal may have more difficulties if ruminations are characterised by repetitive thinking about the interpersonal impact of trauma. In the socio-interpersonal framework of PTSD, trauma is posited have social and contextual impacts on an individual, ranging from feelings of shame and guilt (linked to social connectedness), to disrupting one's sense of belonging with close others or a group (Maercker and Horn, 2013). Moreover, Ehlers and Clark (2000) underscore the that the sense of current threat associated with trauma memories can lead to maladaptive appraisals and ongoing feelings that one's relationships with others have permanently changed for the worse. Furthermore, both PTSD symptoms and rumination are associated with lower social support (Flynn et al., 2010; Kaniasty and Norris, 2008). Since for those high in interdependent self-construal, where interpersonal relationships are central to definitions of the self, thinking repetitively about these interpersonal impacts of trauma may produce greater distress. However, research that is focused on the qualitative content and themes of rumination (particularly those concerning the interpersonal impact of trauma) is needed in order to better understand this interaction effect.

There was no evidence that independent self-construal or holistic thinking style moderated the association between rumination and PTSD symptoms. de Vaus et al. (2018) state that while members of holistic cultures have enhanced capacity to downregulate negative emotional responses, this may not always be predictive of fewer negative emotions. This may reflect the assumption that successful emotional processing of traumatic events requires original emotions to be activated and thus open to change (Foa and Kozak, 1986; Teasdale, 1999) – such activation being less frequent when reactions are downregulated. Additionally, the Holistic Cognition Scale showed limited reliability in this study (α=0.60–0.69) and lacks cross-cultural validation in trauma populations, potentially affecting our ability to detect moderation effects. Future studies should consider an alternate measure of cognitive style. We are not certain why independent self-construal did not interact with the relationship between rumination and PTSD symptoms. It is possible that for trauma survivors with PTSD symptoms, the strong effect of rumination trumps any effect of independent cultural values.

There are several theoretical implications. If these findings can be replicated in future cross-cultural research, existing clinical models which implicate rumination in the maintenance of PTSD may be applicable to Chinese trauma survivors residing in Australia. Second, our findings provide some support for recent cross-cultural accounts that cultural differences in self-construal may influence rumination (de Vaus et al., 2018; Grossman and Kross, 2010). Our findings also support emerging research (e.g., Ford and Mauss, 2015) highlighting the importance of considering an individual's cultural values and self-construal, in addition to cultural group, when conducting cross-cultural clinical research. Beyond group membership, it may be more meaningful to examine how particular processes (such as rumination) interact with an individual's cultural norms and values. Despite this, it is important to note that currently models of PTSD that implicate rumination as a maintenance factor (e.g., Brewin et al., 2003; Ehlers and Clark, 2000) do not include culture as a variable. Thus, there is a need for theoretical accounts of rumination to integrate cross-cultural psychology theories and empirical research to better conceptualize rumination and its role in maintaining psychopathology in Asian contexts.

Clinically, our results highlight the importance of considering rumination regardless of the cultural background of trauma survivors. Thus, given the importance of rumination, in maintaining PTSD, it is essential that formulations and treatment approaches consider the role of rumination in all clients. PTSD treatment approaches would benefit from not only focusing on brooding but also considering trauma-specific rumination (Moulds et al., 2020). Thus, our findings indicate that rumination may be a pan-culturally detrimental mechanism maintaining trauma symptomatology. There was some evidence that interdependent self-construal, regardless of one's cultural background, may influence the association between rumination and PTSD symptoms. Thus, it may be clinically beneficial to consider the impacts of trauma and rumination on interdependent aspects of self, particularly as threats to the self maintains PTSD symptoms (Ehlers and Clark, 2000). Nevertheless, it is also important to consider the socio-cultural context (including an individual's values and self-concept) in assessment, formulation and treatment of PTSD. This will aid in making treatments more culturally sensitive and tailored, which has been shown to significantly improve treatment effects (Huey and Tilley, 2018).

This study had some limitations. First, the cross-sectional design precludes causal and directional inferences about the relationship between rumination and PTSD symptoms. While we controlled for trauma type and time since trauma, unmeasured aspects of trauma severity (e.g., perceived life threat, injury severity) may still influence results. Future longitudinal or experimental designs would help clarify directionality while better accounting for trauma characteristics. Second, as trauma-related questions (e.g., LEC-5 and PCL-5) were presented prior to rumination scales, participants may have been primed to ruminate more on their traumatic experiences. This sequence could inflate correlations between rumination and PTSD symptoms across both groups. Third, although over 30 % of each cultural group reported a PCL-5 score indicative of probable PTSD (Weathers et al., 2013), our community sample may not generalise to a clinical population. Future research should seek to replicate our findings in a trauma survivors with formal PTSD diagnoses. Fourth, this study sampled Chinese Australians and cannot be generalised to other East-Asian groups or Chinese in China. While we restricted our Chinese group to $\leq$ 15 years residence in Australia, some acculturation effects (e.g., emotion expression norms, understanding of trauma) may remain. Future studies could build on our findings by examining these alongside core cultural dimensions such as self-construal and thinking style. Finally, there were significantly more women than men in our study. Although we controlled for gender differences between groups, mean levels of rumination may be inflated by the gender imbalance given that women tend to ruminate more frequently than men (Salk et al., 2017).

In conclusion, rumination was found to relate to PTSD symptoms across European Australian and Chinese trauma survivors, with no evidence that cultural group moderated this relationship. Interdependent self-construal moderated the association between rumination and PTSD, such that the strength of the association between each type of rumination and PTSD symptomatology increased at higher levels of interdependence. A novel finding was that there was no evidence that cultural group and self-construal or holistic cognitive style interacted to impact the association between rumination and PTSD symptoms. This study provides novel insights into the importance of rumination in PTSD, regardless of cultural background.

Author note

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CRediT authorship contribution statement

James Haoxiang Li: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Larissa Shiying Qiu:** Writing – review & editing, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Joshua Wong:** Writing – review & editing, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Winnie Lau:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Richard A. Bryant:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **July Lies:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Belinda J. Liddell:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Laura Jobson:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Laura Jobson reports financial support was provided by National Health and Medical Research Council. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jadr.2025.100924.

References

Austin, P.C., 2011. An introduction to propensity score methods for reducing the effects of confounding in observational studies. Multivariate Behav. Res. 46 (3), 399–424. https://doi.org/10.1080/00273171.2011.568786.

Beaumont, J.F., Rivest, L.P., 2009. Dealing with outliers in survey data. In: Handbook of Statistics, 29. Elsevier Science & Technology, pp. 247–279. https://doi.org/10.1016/S0169-7161(08)00011-4.

Brewin, C.R., Holmes, E.A., 2003. Psychological theories of posttraumatic stress disorder. Clin. Psychol. Rev. 23 (3), 339. https://doi.org/10.1016/S0272-7358(03)00033-3.

Cox, S., Funasaki, K., Smith, L., Mezulis, A.H., 2012. A prospective study of brooding and reflection as moderators of the relationship between stress and depressive symptoms in adolescence. Cognit. Ther. Res. 36 (4), 290–299. https://doi.org/10.1007/s10608-011-9373-z.

Chang, E.C., Tsai, W., Sanna, L.J., 2010. Examining the relations between rumination and adjustment. Asian Am. J. Psychol. 1 (1), 46–56. https://doi.org/10.1037/a0018821.

de Vaus, J., Hornsey, M.J., Kuppens, P., Bastian, B., 2018. Exploring the east-west divide in prevalence of affective disorder: a case for cultural differences in coping with negative emotion. Pers. Soc. Psychol. Rev. 22 (3), 285–304. https://doi.org/10.1177/1088868317736222.

Ehlers, A., Clark, D.M., 2000. A cognitive model of posttraumatic stress disorder. Behav. Res. Ther. 38 (4), 319–345. https://doi.org/10.1016/S0005- 7967(99)00123-0.

Feng, J., Wang, Y., Ji, Z., Zhang, D., 2022. The effect of ostracism on adults' materialism: the roles of security and self-construal. Front. Psychol. 13, 796924. https://doi.org/10.3389/fpsyg.2022.796924. –796924.

Flynn, M., Kecmanovic, J., Alloy, L.B., 2010. An examination of integrated cognitive-interpersonal vulnerability to depression: the role of rumination, perceived social support, and interpersonal stress generation. Cogn. Ther. Res. 34, 456–466. https://doi.org/10.1007/s10608-010-9300-8.

Foa, E.B., Kozak, M.J., 1986. Emotional processing of fear: exposure to corrective information. Psychol. Bull. 99, 20–35. https://doi.org/10.1037/0033-2909.99.1.20.

Ford, B.Q., Mauss, I.B., 2015. Culture and emotion regulation. Curr. Opin. Psychol. 3, 1–5. https://doi.org/10.1016/j.copsyc.2014.12.004.

Fung, H.W., Chan, C., Lee, C.Y., Ross, C.A., 2019. Using the Post-traumatic stress disorder (PTSD) checklist for DSM-5 to screen for PTSD in the Chinese context: a pilot study in a psychiatric sample. J. Evid. Based. Soc. Work 16 (6), 643–651. https://doi.org/10.1080/26408066.2019.1676858 (2019).

Gorecki, C., Brown, J., Briggs, M., Coleman, S., 2014. Language translations & cross-cultural adaptation guideline. NHS Trust. https://ctru.leeds.ac.uk/wp-content/uploads/2019/01/Translation_Guidelines_10JAN14_finalv1.0.pdf.

Grossmann, I., Kross, E., 2010. The impact of culture on adaptive versus maladaptive self-reflection. Psychol. Sci. 21 (8), 1150–1157. https://doi.org/10.1177/0956797610376655.

Harrington, R., Loffredo, D.A., 2011. Insight, rumination, and self-reflection as predictors of well-being. J. Psychol. 145 (1), 39–57. https://doi.org/10.1080/00223980.2010.528072.

He, J., Liu, Y., Cheng, C., Fang, S., Wang, X., Yao, S., 2021. Psychometric properties of the Chinese version of the 10-item ruminative response scale among undergraduates and depressive patients. Front. Psychiatry 12, 626859. https://doi.org/10.3389/fpsyt.2021.626859. –626859.

Hernández-Posadas, A., Lommen, M.J.J., de la Rosa Gómez, A., 2024. Transdiagnostic factors in symptoms of depression and post-traumatic stress: a systematic review. Curr. Psychol. 43, 5933–5948. https://doi.org/10.1007/s12144-023-04792-x.

Hinton, D.E., Lewis-Fernández, R., 2011. The cross-cultural validity of posttraumatic stress disorder: implications for DSM-5. Depress. Anxiety 28 (9), 783–801. https://doi.org/10.1002/da.20753.

Huey, S., Tilley, J., 2018. Effects of mental health interventions with Asian Americans: a review and meta-analysis. J. Consult. Clin. Psychol. 86, 915–930. https://doi.org/10.1037/ccp0000346.

Itoh, M., Ujiie, Y., Nagae, N., Niwa, M., Kamo, T., Lin, M., Hirohata, S., Kim, Y., 2017. A new short version of the Posttraumatic Diagnostic Scale: validity among Japanese adults with and without PTSD. Eur. J. Psychotraumatol. 8 (1), 1364119. https://doi.org/10.1080/20008198.2017.1364119. –10.

Jobson, L., Willoughby, C., Specker, P., Wong, J., Draganidis, A., Lau, W., Liddell, B., 2022. Investigating the associations between cognitive appraisals, emotion regulation and symptoms of posttraumatic stress disorder among Asian American and European American trauma survivors. Sci. Rep. 12 (1), 18127. https://doi.org/10.1038/s41598-022-22995-3. –18127.

Kaniasty, K., Norris, F.H., 2008. Longitudinal linkages between perceived social support and posttraumatic stress symptoms: sequential roles of social causation and social selection. J. Trauma Stress 21 (3), 274–281. https://doi.org/10.1002/jts.20334.

Kwon, H., Yoon, K.L., Kwon, J., 2013. Cultural and gender differences in emotion regulation: relation to depression. Cogn. Emot. 27 (5), 769–782. https://doi.org/10.1080/02699931.2013.792244.

Lux A.A., Grover S.L. & Teo S.T.T. (2021). Development and validation of the holistic cognition scale. Sec. Cultural Psychology, 12. 10.3389/fpsyg.2021.551623.

Mancenido M., Salehi P., Chiou E., Mosallanezhad A., Shah A., & Cohen M. (2021). Challenges of data collection on Mturk: a Human-AI joint face-matching task. IISE Annual Conference. Proceedings, 175–180.

Maercker, A., Horn, A.B., 2013. A socio-interpersonal perspective on PTSD: the case for environments and interpersonal processes. Clin. Psychol. Psychother. 20 (6), 465–481. https://doi.org/10.1002/cpp.1805.

Marjanovic, Z., Struthers, C.W., Cribbie, R., Greenglass, E.R., 2014. The Conscientious responders scale. Sage Open 4 (3), 215824401454596. https://doi.org/10.1177/2158244014545964.

Markus, H., Kitayama, S., 2010. Cultures and selves: a cycle of mutual constitution. Perspect. Psychol. Sci. 5 (4), 420–430. https://doi.org/10.1177/1745691610375557.

McEvoy, P.M., Mahoney, A.E.J., Moulds, M.L., 2010. Are worry, rumination, and post-event processing one and the same? Development of the repetitive thinking questionnaire. J. Anxiety Disord. 24, 509–519. https://doi.org/10.1016/j.janxdis.2010.03.008.

Moulds, M.L., Bisby, M.A., Wild, J., Bryant, R.A., 2020. Rumination in posttraumatic stress disorder: a systematic review. Clin. Psychol. Rev. 82, 101910. https://doi.org/10.1016/j.cpr.2020.101910. –101910.

Nagulendran, A., Jobson, L., 2020. Exploring cultural differences in the use of emotion regulation strategies in posttraumatic stress disorder. Eur. J. Psychotraumatol. 11 (1), 172–179. https://doi.org/10.1080/20008198.2020.1729033.

Nisbett, R.E., Peng, K., Choi, I., Norenzayan, A., 2010. Culture and systems of thought: holistic versus analytic cognition. Psychol. Rev. 108, 291–310. https://doi.org/10.1037//0033-295X.108.2.291.

Nolen-Hoeksema, S., 1991. Responses to depression and their effects on the duration of depressive episodes. J. Abnorm. Psychol. 100 (4), 569–582. https://doi.org/10.1037/0021-843X.100.4.569 (1965).

Nolen-Hoeksema, S., Morrow, J., 1991. A prospective study of depression and posttraumatic stress symptoms after a natural disaster: the 1989 Loma Prieta Earthquake. J. Pers. Soc. Psychol. 61 (1), 115–121. https://doi.org/10.1037/0022-3514.61.1.115.

Salk, R.H., Hyde, J.S., Abramson, L.Y., 2017. Gender differences in depression in representative national samples: meta-analyses of diagnoses and symptoms. Psychol. Bull. 143 (8), 783–822. https://doi.org/10.1037/bul0000102.

Seligowski, A.V., Lee, D.J., Bardeen, J.R., Orcutt, H.K., 2015. Emotion regulation and posttraumatic stress symptoms: a meta-analysis. Cogn. Behav. Ther. 44 (2), 87–102. https://doi.org/10.1080/16506073.2014.980753.

Schwartz, S.J., Unger, J.B., Zamboanga, B.L., Szapocznik, J., 2010. Rethinking the concept of acculturation: implications for theory and research. Am. Psychol. 65 (4), 237–251. https://doi.org/10.1037/a0019330.

Singelis, T.M., 1994. The measurement of independent and interdependent self-construals. Pers. Soc. Psychol. Bull. 20, 580–591. https://doi.org/10.1177/0146167294205014.

Stuart, E.A., 2010. Matching methods for causal inference: a review and a look forward. Stat. Sci. 25 (1), 1–21. https://doi.org/10.1214/09-sts313.

Teasdale, J.D., 1999. Emotional processing, three modes of mind and the prevention of relapse in depression. Behav. Res. Ther. 37, S53–S77. https://doi.org/10.1016/S0005-7967(99)00050-9.

Toussaint, L., Lee, J.A., Hyun, M.H., Shields, G.S., Slavich, G.M., 2023. Forgiveness, rumination, and depression in the United States and Korea: a cross-cultural mediation study. J. Clin. Psychol. 79 (1), 143–157. https://doi.org/10.1002/jclp.23376.

Treynor, W., Gonzalez, R., Nolen-Hoeksema, S., 2003. Rumination reconsidered: a psychometric analysis. Cognit. Ther. Res. 27 (3), 247–259. https://doi.org/10.1023/A:1023910315561.

Weathers F.W., Litz B.T., Keane T.M., Palmieri P.A., Marx B.P., & Schnurr P.P. (2013). *The PTSD Checklist for DSM–5 (PCL-5)*. http://www.ptsd.va.gov/professional/assessment/adult-sr/ptsd-checklist.asp.

Wei, X., Shen, P., Li, T., Liu, X., 2022. Self-construal mediates the impact of job burnout on life satisfaction and Chinese happiness. Curr. Psychol. 41 (4), 2012–2023. https://doi.org/10.1007/s12144-020-00735-y.

Wong, S.M.Y., Chen, E.Y.H., Lee, M.C.Y., Suen, Y.N., Hui, C.L.M., 2023. Rumination as a transdiagnostic phenomenon in the 21st century: the flow model of rumination. Brain Sci. 13 (7), 1041–1044. https://doi.org/10.3390/brainsci13071041.

Zhang, M., Cross, S.E., 2011. Emotions in memories of success and failure: a cultural perspective. Emotion 11, 866–880. https://doi.org/10.1037/a002402.